

# WENYOU (TOBIAS) TIAN

Year 4, Honours in Computer Science, Major in Economics

+1-365-228-7285   [wtian02@student.ubc.ca](mailto:wtian02@student.ubc.ca)   [vansdardy.github.io](https://vansdardy.github.io)   [github.com/vansdardy](https://github.com/vansdardy)   [linkedin.com/in/wy-t-tian](https://linkedin.com/in/wy-t-tian)

## Education

### The University of British Columbia

Sep. 2022 – Jun. 2028

*Honours Bachelor of Science in Computer Science, Bachelor of Arts in Economics*

*Vancouver, BC, Canada*

- Algorithm Design and Analysis; Software Engineering; Computer Systems; Computer Network; Quantum Computing
- Advanced Econometrics; Game Theory; Money and Banking; Micro/Macroeconomic Analysis; Intl. Finance/Trade

### ETH Zürich - Department of Computer Science

Sep. 2024 – Feb. 2025

*Exchange Student*

*Zürich, Switzerland*

- Algorithms, Probability, Computing; Compiler Design; Computer Systems (OS + DS); Quantum Physics

## Technical Skills

**Computer Languages:** Java, Python, C/C++, R, HTML/CSS/JavaScript, OCaml, TypeScript, SQL (basic)

**Developer Tools:** Git, VS Code, IntelliJ IDEA, JupyterLab, IDLE, Docker

**Technologies/Frameworks:** Matplotlib, pandas, NumPy, SymPy, Tidyverse, lubridate, haven, Node.js, Express

**Research Methodologies:** OLS regression, Hypothesis Testing, RAG Pipelines, Agentic AI systems, Data Cleaning

## Work Experience

### Software Research Engineer

Jan. 2026 - Dec. 2026

*Noah's Ark Lab - Huawei Canada | Python, TypeScript, Git, Node.js, Agentic AI*

*Markham, Ontario*

- Led the design and implementation of a knowledge base harness, which is a TypeScript OpenCode plugin for building large-scale knowledge bases, using a "deterministic kernel + LLM semantic agents" architecture aiming to process 3,000+ Markdown documents, with checkpoint/resume and incremental update support.
- Prototyped a tool for converting XML-based UI metadata into hierarchical UI trees, processing 700+ screenshots to support GUI agent interaction (project later transitioned across teams).

### Undergraduate Academic Assistant

Jun. 2024 - Present

*Science One Physics, Year 1 Physics | Python, HTML/Markdown, Git, SymPy*

*University of British Columbia*

- Reformatted and randomized more than 350 freshman-level physics problems with HTML and Markdown for frontend format display, and Python for backend question generation and randomization that assists 6+ physics professors to set up quizzes and exams, reaching hundreds of freshman students each year.
- Integrated the PrairieLearn autograder system to correctly identify and grade mathematical expressions that involve alphabetical variables and physical units using the SymPy library API which has served 151 students over two years.

### Undergraduate Teaching Assistant (Mathematics and Physics)

Sep. 2023 – Apr. 2024

*Science One Program*

*The University of British Columbia*

- Held bi-weekly office hours and assisted in organizing tutorials and workshops that answered questions regarding freshman-level mathematics and physics, including calculus, classical and quantum mechanics, and special relativity.
- Coded short and simple Python programs to demonstrate mathematical and physical concepts numerically, such as Riemann sum approximation and Euler's method approximation.

### Circulation and Shelving Assistant

Sep. 2023 – Apr. 2024

*Law Library*

*The University of British Columbia*

- Organized and rearranged books, journals, law statutes etc., by the Library of Congress Call Number System, across about 500 shelves each semester.
- Maintained book circulation within and across campuses, including assisting patrons with requested book information and personal inquiries, including answering phone calls.

## Projects

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### Horizon Holdings | *Python, SQLite, Electron, Claude Code*

Aug. 2026

- Specified and directed the build of a cross-platform portfolio tracker, whose code written entirely using Claude Code, modeling a self-designed 78-company, 12-market, 7-currency index against real brokerage holdings, in daily personal use for at least a month.
- Diagnosed and directed fixes for a currency-representation defect (LSE securities quote in GBX, not GBP) and a cross-timezone data-staleness gap between same-day US pricing and T+1 European/Japanese feeds.
- Set a data-integrity-first requirement where the system must refuse to record a portfolio value when any constituent price is missing, rather than defaulting silently

### Insight UBC | *TypeScript, HTML/CSS/JavaScript, chai-as-promised, Mocha, Express, JSON, VS Code, Git*

Apr. 2025

- Collaborated with a peer to develop a full-stack web application where users can upload and add, remove existing, list existing UBC's course section datasets, and perform queries on them in JSON-formatted files compressed as one zip file.
- Combined using chai-as-promised for smoke tests, unit tests, integration tests, and end-to-end tests that ensure the functionality of each data structure and the entire application.
- Utilized Express for the web application's framework to write RESTful APIs which allow sending requests and receiving responses with proper and reasonable status codes and results or errors.

### A Research on UBC Dining Halls' Food Waste | *R, Tidyverse, Econometric Analysis*

Apr. 2024

- Applied multivariate OLS regression to 270+ observations across three dining halls to quantify the impact of meal-period foot traffic on daily food waste.
- Diagnosed heteroskedasticity using White's test and re-estimated models with heteroskedasticity-consistent (HC) robust standard errors; performed hypothesis testing, including F-tests, by meal period.
- Identified statistically significant effects at Orchard Commons Dining Hall (breakfast  $\beta = -0.25$ , lunch  $\beta = +0.37$ , both  $p < 0.01$ ) and validated results through robustness checks using simplified model specifications.

### Oat Compiler | *OCaml, VS Code, Docker, Linux*

Dec. 2024

- Developed a complete compiler for the Oat language, encompassing an x86lite simulator, LLVMlite-based IR construction, type checking, and simple optimizations over a four-month period.
- Designed and implemented a register allocator using Kempe's K-coloring algorithm to optimize register utilization compared to the previous spill-to-stack approach.
- Implemented core Oat language features, including arrays, functions, and control flow constructs, while validating correctness through systematic testing of each compilation phase.

### Classical Enigma Machine | *Java, JUnit, JSON, Swing, IntelliJ IDEA*

Dec. 2023

- Digitalized a German cryptographic device used during WWII, and expanded the functionality to incorporate more input/output characters subject to user customization.
- Utilized JUnit testing library to ensure the correctness of each class method, and guaranteed data persistence with JSON for users to load or save at each session.
- Designed a GUI using the Swing library in which users can interact with the functions of the program.

### Earth's Albedo via Relative Photometry | *Python, NumPy, Matplotlib, Mathematical modelling*

Apr. 2023

- Derived and implemented a photometric Earth albedo model from primary literature, incorporating earthshine ratios, lunar phase functions, and atmospheric extinction corrections.
- Processed telescope image data using AstroImageJ and performed Python-based analyses across ISO and exposure parameter space; estimated Earth albedo at 0.262 (vs. literature value of 0.30).
- Presented results at a student-led academic conference and co-authored a 29-page research paper with a faculty mentor in the Department of Physics and Astronomy at UBC.

## Scholarships and Awards

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### J Fred Muir Memorial Scholarship in Science

Mar. 2026

Value: CA\$ 210

*The University of British Columbia*

- Awarded to students in the Faculty of Science on the recommendation of the Faculty for academic excellence.

### Hugh M Brock Go Global Award

Sep. 2024

Value: CA\$ 1,500

*The University of British Columbia*

- Award to UBC students participating in international learning programs through Go Global.